

Aim

To identify a real-world problem, analyze its feasibility, and prepare a project proposal for the development of a MEAN/MERN stack web application.

Project Title

TransitOps – Smart Transport Operations Platform

Problem Statement

Many logistics and transport companies still rely on spreadsheets, paper records, and manual communication to manage their fleets. This results in scheduling conflicts, underutilized vehicles, expired driver licenses, delayed maintenance, inaccurate expense tracking, and poor operational visibility.

A centralized web application is required to automate transport operations by managing vehicles, drivers, trips, maintenance, fuel expenses, and reports in one platform.

Objectives

- Develop a centralized transport management platform.
 - Digitize vehicle registration, vehicle insurance and tracking.
 - Manage driver information and license validity.
 - Automate trip assignment.
 - Prevent invalid dispatches through business rules.
 - Track maintenance history.
 - Record fuel and operational expenses.
 - Generate analytical reports and dashboards.
 - Improve operational efficiency.
 - Reduce manual paperwork.
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Existing System

Current logistics companies generally use:

- Excel spreadsheets
- Manual logbooks
- Phone calls
- WhatsApp communication
- Paper maintenance records
- Manual expense calculations

Problems:

- Duplicate vehicle assignments
 - Driver scheduling conflicts
 - Missed maintenance
 - Expired licenses
 - Manual calculations
 - Data inconsistency
 - Poor visibility
 - Time-consuming reporting
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Proposed System

TransitOps is a centralized MERN Stack web application that manages:

- Vehicle Registry
- Driver Management
- Trip Management
- Maintenance Tracking
- Fuel Logs
- Expense Management
- Dashboard
- Analytics
- Role-Based Access Control

The system automatically validates business rules before dispatching vehicles and updates vehicle/driver status throughout the trip lifecycle

Scope

The project can be used by:

- Logistics companies
- Courier services
- School bus operators
- Fleet operators
- Government transport departments
- Construction companies
- Rental vehicle businesses

Future scope includes:

- GPS Tracking
- Route Optimization
- Mobile App
- AI-based Predictive Maintenance
- IoT Vehicle Monitoring

Expected Users

User	Responsibility
Fleet Manager	Manage vehicles and maintenance
Dispatcher	Create and assign trips
Driver	View assigned trips
Safety Officer	Monitor licenses and safety
Financial Analyst	Track expenses and profitability
Administrator	Manage users and permissions

Expected Features

Authentication

- Login
- Logout
- Role-Based Access Control (RBAC)

Dashboard

- Active Vehicles
- Available Vehicles
- Maintenance Vehicles
- Fleet Utilization
- Active Trips

Vehicle Management

- Add Vehicle
- Update Vehicle
- Delete Vehicle
- Vehicle Status

Driver Management

- Driver Profile
- License Expiry
- Safety Score

Trip Management

- Create Trip
- Assign Driver
- Assign Vehicle
- Complete Trip
- Cancel Trip

Maintenance

- Maintenance Records
- Automatic Status Update

Fuel & Expense

- Fuel Logs
- Maintenance Cost
- Toll Expenses

Reports

- Fuel Efficiency
- Operational Cost
- Vehicle Return on Investment (ROI)
- Fleet Utilization
- CSV Export
- PDF Export

Extra Features

- Email Reminders
 - Dark Mode
 - Search & Filters
 - Charts
 - Vehicle Documents
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Technology Stack

Frontend

- React.js
- HTML5
- CSS3
- JavaScript

Backend

- Node.js
- Express.js

Database

- MongoDB

Authentication

- JWT (JSON Web Token)
- bcrypt (Password Encryption)

Analytics Dashboard

- Chart.js

Document Upload

- Cloudinary

Development Tools

- Visual Studio Code
- Postman
- npm

Version Control

- Git
 - GitHub
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